Derivatives of some Activation Functions

$$\begin{cases} 0 & \text{for } x<0 \\ 1 & \text{for } x\ge 0 \end{cases}$$

Derivative $\Rightarrow$

$$\begin{cases} 0 & \text{for } x\neq 0 \\ ? & \text{for } x=0 \end{cases}$$

$$\begin{cases} -1 & \text{for } x<0 \\ 1 & \text{for } x\ge 0 \end{cases}$$

Derivative $\Rightarrow$

$$\begin{cases} 0 & \text{for } x\neq 0 \\ ? & \text{for } x=0 \end{cases}$$

$$f(x) = \text{Sigmoid}(x) = \frac{1}{1+e^{-x}}$$

Derivative $\Rightarrow$

$$f'(x) = f(x)\,(1 - f(x))$$

$$f(x) = \tanh(x) = \frac{\sinh x}{\cosh x}$$

Derivative $\Rightarrow$

$$f'(x) = 1 - f(x)^2$$

$$f(x) = \tan^{-1}(x)$$

Derivative $\Rightarrow$

$$f'(x) = \frac{1}{x^2+1}$$